given page in the flip book of spacetime. But when Einstein realized that two observers in relative motion have clocks that tick off time differently, this all changed. Clocks that are moving relative to each other fall out of synchronization and therefore give different notions of simultaneity. Each page in Figure 3.3b is but one observer's view of the events in space taking place at a given moment of his or her time. Another observer, moving relative to the first, will declare that the events on a single one of these pages *do not* all happen at the same time.

This is known as the *relativity of simultaneity,* and we can see it directly. Imagine that Itchy and Scratchy, pistols in paws, are now facing each other on opposite ends of a long, moving railway car with one referee on the train and another officiating from the platform. To make the duel as fair as possible, all parties have agreed to forgo the three-step rule, and instead, the duelers will draw when a small pile of gunpowder, set midway between them, explodes. The first referee, Apu, lights the fuse, takes a sip of his refreshing Chutney Squishee, and steps back. The gunpowder flares, and both Itchy and Scratchy draw and fire. Since Itchy and Scratchy are the same distance from the gunpowder, Apu is certain that light from the flare reaches them simultaneously, so he raises the green flag and declares it a fair draw. But the second referee, Martin, who was watching from the platform, wildly squeals foul play, claiming that Itchy got the light signal from the explosion before Scratchy did. He explains that because the train was moving forward, Itchy was heading toward the light while Scratchy was moving away from it. This means that the light did not have to travel quite as far to reach Itchy, since he moved closer to it; moreover, the light had to travel farther to reach Scratchy, since he moved away from it. Since the speed of light, moving left or right from anyone's perspective, is constant, Martin claims that it took the light longer to reach Scratchy since it had to travel farther, rendering the duel unfair.

Who is right, Apu or Martin? Einstein's unexpected answer is that they both are. Although the conclusions of our two referees differ, the observations and the reasoning of each are flawless. Like the bat and the baseball, they simply have different perspectives on the same sequence of events. The shocking thing that Einstein revealed is that their different perspectives yield different but equally valid claims of what events happen at the same time. Of course, at everyday speeds like that of the